SPDM Update:
Hybrid, Authorization Policy and Standard Signatures (FRU)

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Disclaimer

- The information in this presentation represents a snapshot of work in progress within the DMTF SPDM WG.
- This information is subject to change without notice. The standard specifications remain the normative reference for all information.
- For additional information, see the DMTF website.
- This information is a summary of the information that will appear in the specifications. See the specifications for further details.

What is SPDM?

- The DMTF's Security Protocols and Data Models (SPDM) Working group is responsible for the SPDM suite of specifications.
- These standards enable authentication, attestation and key exchange to assist in enabling infrastructure security.
- Other specifications define enhancements to infrastructure security through additional functionality or mapping SPDM to additional transports.



SPDM Feature Summary

- Version 1.0:
  - Measurements
  - Device Attestation and Authentication
- Version 1.1:
  - Secure Session
    - Public Key Exchange
    - Symmetric Key Exchange
  - Mutual Authentication
- Version 1.2:
  - Installation of certificates
  - Allows for alias certificates derived from device certificates
  - Send and receive large SPDM messages (chunks)
  - Added SM2, SM3, SM4 algorithms to supported list
  - New OIDs added
  - Deprecated basic mutual authentication in CHALLENGE and CHALLENGE_AUTH
- Version 1.3
  - Eventing
  - Multi-Key
  - Generic Certificates
  - MEL & HEM
  - Endpoint Info
- Version 1.4
  - PQC Support
  - Certificate Slot Management Control
- Authorization
  - Authorization Policy
- Bindings
  - MCTP to SPDM
  - MCTP to Secure Messages
  - Secure Messages to SPDM
  - SPDM over TCP
  - SPDM over Storage (NVMe, SAS, SATA)

Hybrid for SPDM 1.5

Requirements:

1. Minimal changes to the SPDM 1.4 message flow.
2. SPDM 1.4 devices with a Traditional certificate chain and a PQC certificate chain are upgradeable to SPDM 1.5 with hybrid support.
3. Hybrid algorithm combinations are defined in SPDM 1.5. i.e., endpoints cannot “mix-and-match” or create arbitrary combinations. Keep number of supported hybrid *combinations to a minimum*.
4. Minimal divergence from TLS.

Note:

- There are other changes coming in SPDM 1.5, such as new measurement details. There is insufficient time to cover these. Please see the WIP.

Algorithm Combinations

Hybrid Signature

1. MLDSA87-ECDSA-P384
2. MLDSA87-Ed448
3. MLDSA87-RSA3072-PSS
4. MLDSA87-RSA4096-PSS

Key Exchange

1. MLKEM1024-ECDH-P384
2. MLKEM1024-X448
3. MLKEM1024-ECDH-P521

- Please provide feedback on any other combinations that may be required

Attestation

- Based on the OCP request for GET_ATTESTATION support
  - Uses a self-contained signature to enable remote verification
- Summary of changes
  - Discovery of EndpointInfo Sub-Codes
  - Discovery of supported attestation profiles
  - Read of an attestation Claim based on profile

Authorization Policy

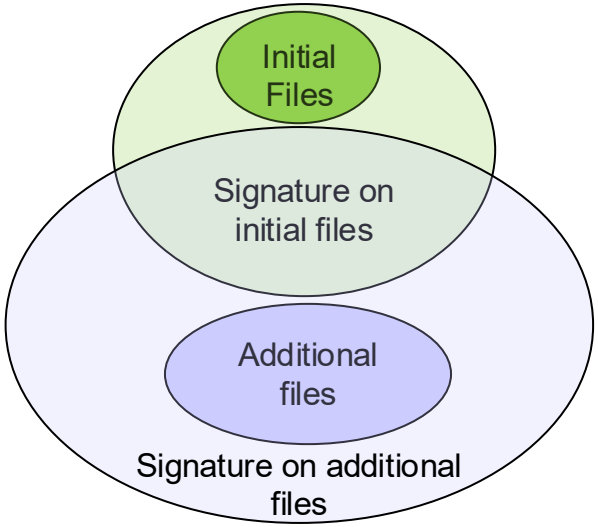
- DSP0289 – SPDM Authorization Specification (Dec 2025)
  - Authorization is a general mechanism for any use case in platform management to ensure if the requesting entity has the appropriate privileges to perform protected actions.
  - Policy is associated with the privilege
- DSP0294 – DMTF Authorization Policy Spec (Aug 2026)
  - Rather than create separate specs, DMTF created a single spec for authorization of actions in other DMTF specs.
  - This spec covers SPDM & PLDM FW update
  - Other specifications will be added as use cases are brought forward

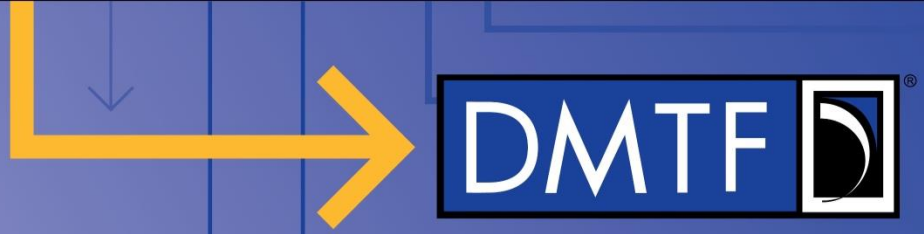
Standard Signatures

- Several spec projects in DMTF need signed objects
  - Signing is not specific to the data being signed
  - The industry only needs one way to sign data
- There are many existing signing formats, most are an envelope structure
  - DMTF has use cases that require retrofitting signing in existing data structures, such as FRU
  - While SPDM creates the signing format, other workgroups will also reference it
  - Need to support traditional and PQC

Signature in FRU

- Signature would be just another file in the file table
- Small header would dictate which files in the file table are covered by the signature
- Different parts of the FRU are laid down at different parts of the manufacturing process
  - Potentially by different organizations
  - Can't just do a single signature at the end covering everything
  - Plan to include a method to sign each extension
  - Plan to include a way to sign the file table and IPMI FRU





For more information,
visit dmtf.org

Learn about the SPDM working group at
dmtf.org/standards/spdm

Thank you!